

EXPERIMENT-3: Client-side validation with JavaScript

Use JavaScript for doing client-side validation of the pages implemented in experiments 1 and 2.

Objective: Add form validation, user feedback, and field rules in the frontend.

Tools : JavaScript, DOM, form validation

Outcome : Validated registration, login, and checkout forms with improved UX.

Answer

Add event listeners for form submission, validate required fields and email format, and display inline validation feedback before submission.

File Name: index.html

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Week 3 - Client-Side Form Validation</title>
    <link
      href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
      rel="stylesheet"
    />
    <link rel="stylesheet" href="styles.css" />
  </head>
  <body>
    <nav class="navbar navbar-expand-lg navbar-dark bg-dark sticky-top">
      <div class="container">
        <a class="navbar-brand fw-bold" href="#">Node.js Lab</a>
        <button
          class="navbar-toggler"
          type="button"
          data-bs-toggle="collapse"
          data-bs-target="#navbarNav"
          aria-controls="navbarNav"
          aria-expanded="false"
```

```

        aria-label="Toggle navigation"
    >
    <span class="navbar-toggler-icon"></span>
</button>
<div class="collapse navbar-collapse" id="navbarNav">
    <ul class="navbar-nav ms-auto">
        <li class="nav-item"><a class="nav-link active" href="#">Home</a></li>
        <li class="nav-item"><a class="nav-link" href="#forms">Forms</a></li>
    </ul>
</div>
</div>
</nav>

```

```

<main class="container py-4 py-lg-5">
    <section class="hero-card p-4 p-lg-5 rounded-4 text-white mb-4">
        <div class="row align-items-center g-4">
            <div class="col-lg-8">
                <h1 class="display-6 fw-bold mb-3">JavaScript form validation for a better user
experience</h1>
                <p class="mb-0">
                    This page demonstrates required-field checks, email validation, password strength,
and checkout rules with inline feedback.

```

```

                </p>
            </div>
            <div class="col-lg-4 text-lg-end">
                <span class="badge bg-light text-dark fs-6">Experiment 3</span>
            </div>
        </div>
    </section>

```

```

<section id="forms" class="row g-4">
    <div class="col-lg-6">
        <div class="card shadow-sm border-0 h-100">
            <div class="card-body p-4">

```

```
<h2 class="h4 fw-bold mb-3">Registration Form</h2>
<form id="registerForm" novalidate>
  <div class="mb-3">
    <label for="fullName" class="form-label">Full name</label>
    <input type="text" class="form-control" id="fullName" />
    <div class="invalid-feedback"></div>
  </div>

  <div class="mb-3">
    <label for="registerEmail" class="form-label">Email address</label>
    <input type="email" class="form-control" id="registerEmail" />
    <div class="invalid-feedback"></div>
  </div>

  <div class="mb-3">
    <label for="registerPassword" class="form-label">Password</label>
    <input type="password" class="form-control" id="registerPassword" />
    <div class="invalid-feedback"></div>
  </div>

  <div class="mb-3">
    <label for="confirmPassword" class="form-label">Confirm password</label>
    <input type="password" class="form-control" id="confirmPassword" />
    <div class="invalid-feedback"></div>
  </div>

  <button type="submit" class="btn btn-primary w-100">Create Account</button>
  <div class="alert alert-success mt-3 d-none" id="registerMessage"
role="status"></div>
</form>
</div>
</div>
</div>
```

```
<div class="col-lg-6">
  <div class="card shadow-sm border-0 h-100">
    <div class="card-body p-4">
      <h2 class="h4 fw-bold mb-3">Login Form</h2>
      <form id="loginForm" novalidate>
        <div class="mb-3">
          <label for="loginEmail" class="form-label">Email address</label>
          <input type="email" class="form-control" id="loginEmail" />
          <div class="invalid-feedback"></div>
        </div>

        <div class="mb-3">
          <label for="loginPassword" class="form-label">Password</label>
          <input type="password" class="form-control" id="loginPassword" />
          <div class="invalid-feedback"></div>
        </div>

        <div class="form-check mb-3">
          <input class="form-check-input" type="checkbox" id="rememberMe" />
          <label class="form-check-label" for="rememberMe">Remember me</label>
        </div>

        <button type="submit" class="btn btn-outline-primary w-100">Login</button>
        <div class="alert alert-info mt-3 d-none" id="loginMessage" role="status"></div>
      </form>
    </div>
  </div>
</div>
```

```
<div class="col-12">
  <div class="card shadow-sm border-0">
    <div class="card-body p-4">
      <h2 class="h4 fw-bold mb-3">Checkout Form</h2>
```

```
<form id="checkoutForm" novalidate>
  <div class="row g-3">
    <div class="col-md-6">
      <label for="checkoutName" class="form-label">Full name</label>
      <input type="text" class="form-control" id="checkoutName" />
      <div class="invalid-feedback"></div>
    </div>
    <div class="col-md-6">
      <label for="checkoutEmail" class="form-label">Email</label>
      <input type="email" class="form-control" id="checkoutEmail" />
      <div class="invalid-feedback"></div>
    </div>
    <div class="col-12">
      <label for="address" class="form-label">Shipping address</label>
      <textarea class="form-control" id="address" rows="3"></textarea>
      <div class="invalid-feedback"></div>
    </div>
    <div class="col-md-6">
      <label for="phone" class="form-label">Phone number</label>
      <input type="text" class="form-control" id="phone" />
      <div class="invalid-feedback"></div>
    </div>
    <div class="col-md-6">
      <label for="cardNumber" class="form-label">Card number</label>
      <input type="text" class="form-control" id="cardNumber" />
      <div class="invalid-feedback"></div>
    </div>
    <div class="col-md-6">
      <label for="expiry" class="form-label">Expiry (MM/YY)</label>
      <input type="text" class="form-control" id="expiry" placeholder="08/29" />
      <div class="invalid-feedback"></div>
    </div>
    <div class="col-md-6">
```

```

        <label for="cvv" class="form-label">CVV</label>
        <input type="text" class="form-control" id="cvv" />
        <div class="invalid-feedback"></div>
    </div>
</div>

    <button type="submit" class="btn btn-success w-100 mt-4">Place Order</button>
    <div class="alert alert-success mt-3 d-none" id="checkoutMessage"
role="status"></div>
</form>
</div>
</div>
</div>
</section>
</main>

<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"></scrip
t>
    <script src="script.js"></script>
</body>
</html>

```

File Name: styles.css

Code:

```

body {
    background: linear-gradient(135deg, #f8fafc 0%, #eef2ff 100%);
    min-height: 100vh;
}

.hero-card {
    background: linear-gradient(120deg, #2563eb, #4f46e5);
}

```

```
.form-control.is-invalid {  
  border-color: #dc3545;  
}
```

```
.form-control.is-valid {  
  border-color: #198754;  
}
```

```
.invalid-feedback {  
  display: block;  
}
```

File Name: script.js

```
const emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
```

```
function setFeedback(input, message, isValid) {  
  const feedback = input.parentElement.querySelector('.invalid-feedback');  
  input.classList.remove('is-valid', 'is-invalid');
```

```
  if (message) {  
    input.classList.add(isValid ? 'is-valid' : 'is-invalid');  
    feedback.textContent = message;  
  } else {  
    feedback.textContent = "";  
  }  
}
```

```
function validateRequired(input, label) {  
  const value = input.value.trim();  
  if (!value) {
```

```
    setFeedback(input, `${label} is required.`, false);  
    return false;  
  }  
  setFeedback(input, "", true);  
  return true;  
}
```

```
function validateEmail(input, label) {  
  if (!validateRequired(input, label)) return false;  
  
  if (!emailPattern.test(input.value.trim())) {  
    setFeedback(input, `Please enter a valid ${label.toLowerCase()}.`, false);  
    return false;  
  }  
}
```

```
  setFeedback(input, "", true);  
  return true;  
}
```

```
function validatePassword(input) {  
  const value = input.value;  
  if (value.length < 6) {  
    setFeedback(input, 'Password must be at least 6 characters long.', false);  
    return false;  
  }  
  setFeedback(input, "", true);  
  return true;  
}
```

```
function validateConfirmPassword(passwordInput, confirmInput) {  
  if (!validateRequired(confirmInput, 'Confirm password')) return false;  
  if (passwordInput.value !== confirmInput.value) {  
    setFeedback(confirmInput, 'Passwords do not match.', false);  
  }  
}
```



```
    return false;
}
setFeedback(confirmInput, "", true);
return true;
}
```

```
function validatePhone(input) {
    const value = input.value.trim();
    const phonePattern = /^\\d{10}$/;
    if (!phonePattern.test(value)) {
        setFeedback(input, 'Phone number must be 10 digits.', false);
        return false;
    }
    setFeedback(input, "", true);
    return true;
}
```

```
function validateCardNumber(input) {
    const value = input.value.replace(/\\s+/g, "");
    if (!/^\\d{16}$/ .test(value)) {
        setFeedback(input, 'Card number must be 16 digits.', false);
        return false;
    }
    setFeedback(input, "", true);
    return true;
}
```

```
function validateExpiry(input) {
    const value = input.value.trim();
    const expiryPattern = /^(0[1-9]|1[0-2])\\d{2}$/;
    if (!expiryPattern.test(value)) {
        setFeedback(input, 'Use the format MM/YY.', false);
        return false;
    }
}
```

```
}
```

```
const [month, year] = value.split('/').map(Number);  
const currentDate = new Date();  
const currentYear = currentDate.getFullYear() % 100;  
const currentMonth = currentDate.getMonth() + 1;
```

```
if (year < currentYear || (year === currentYear && month < currentMonth)) {  
  setFeedback(input, 'Expiry date has already passed.', false);  
  return false;  
}
```

```
setFeedback(input, "", true);  
return true;  
}
```

```
function validateCvv(input) {  
  const value = input.value.trim();  
  if (!/^\d{3,4}$/.test(value)) {  
    setFeedback(input, 'CVV must be 3 or 4 digits.', false);  
    return false;  
  }  
  setFeedback(input, "", true);  
  return true;  
}
```

```
function handleFormSubmit(form, messageBox, validator) {  
  form.addEventListener('submit', function (event) {  
    event.preventDefault();  
    const isValid = validator();  
    if (!isValid) {  
      messageBox.classList.add('d-none');  
      return;  
    }  
  });  
}
```

```

    }

    messageBox.textContent = 'Form submitted successfully!';
    messageBox.classList.remove('d-none');
    form.reset();
  });
}

function bindRealTimeValidation(inputs, validator) {
  inputs.forEach((input) => {
    input.addEventListener('input', () => validator());
    input.addEventListener('blur', () => validator());
  });
}

const registerForm = document.getElementById('registerForm');
const registerMessage = document.getElementById('registerMessage');

function validateRegisterForm() {
  const fullName = document.getElementById('fullName');
  const email = document.getElementById('registerEmail');
  const password = document.getElementById('registerPassword');
  const confirm = document.getElementById('confirmPassword');

  const checks = [
    validateRequired(fullName, 'Full name'),
    validateEmail(email, 'Email'),
    validatePassword(password),
    validateConfirmPassword(password, confirm),
  ];

  return checks.every(Boolean);
}

```

```
handleFormSubmit(registerForm, registerMessage, validateRegisterForm);
bindRealTimeValidation(
  [
    document.getElementById('fullName'),
    document.getElementById('registerEmail'),
    document.getElementById('registerPassword'),
    document.getElementById('confirmPassword'),
  ],
  validateRegisterForm
);
```

```
const loginForm = document.getElementById('loginForm');
const loginMessage = document.getElementById('loginMessage');
```

```
function validateLoginForm() {
  const email = document.getElementById('loginEmail');
  const password = document.getElementById('loginPassword');

  const checks = [validateEmail(email, 'Email'), validateRequired(password, 'Password')];
  return checks.every(Boolean);
}
```

```
handleFormSubmit(loginForm, loginMessage, validateLoginForm);
bindRealTimeValidation(
  [document.getElementById('loginEmail'), document.getElementById('loginPassword')],
  validateLoginForm
);
```

```
const checkoutForm = document.getElementById('checkoutForm');
const checkoutMessage = document.getElementById('checkoutMessage');
```

```
function validateCheckoutForm() {
```

```
const name = document.getElementById('checkoutName');
const email = document.getElementById('checkoutEmail');
const address = document.getElementById('address');
const phone = document.getElementById('phone');
const cardNumber = document.getElementById('cardNumber');
const expiry = document.getElementById('expiry');
const cvv = document.getElementById('cvv');
```

```
const checks = [
  validateRequired(name, 'Full name'),
  validateEmail(email, 'Email'),
  validateRequired(address, 'Address'),
  validatePhone(phone),
  validateCardNumber(cardNumber),
  validateExpiry(expiry),
  validateCvv(cvv),
];
```

```
return checks.every(Boolean);
}
```

```
handleFormSubmit(checkoutForm, checkoutMessage, validateCheckoutForm);
bindRealTimeValidation(
```

```
[
  document.getElementById('checkoutName'),
  document.getElementById('checkoutEmail'),
  document.getElementById('address'),
  document.getElementById('phone'),
  document.getElementById('cardNumber'),
  document.getElementById('expiry'),
  document.getElementById('cvv'),
],
```

```
validateCheckoutForm
```

);